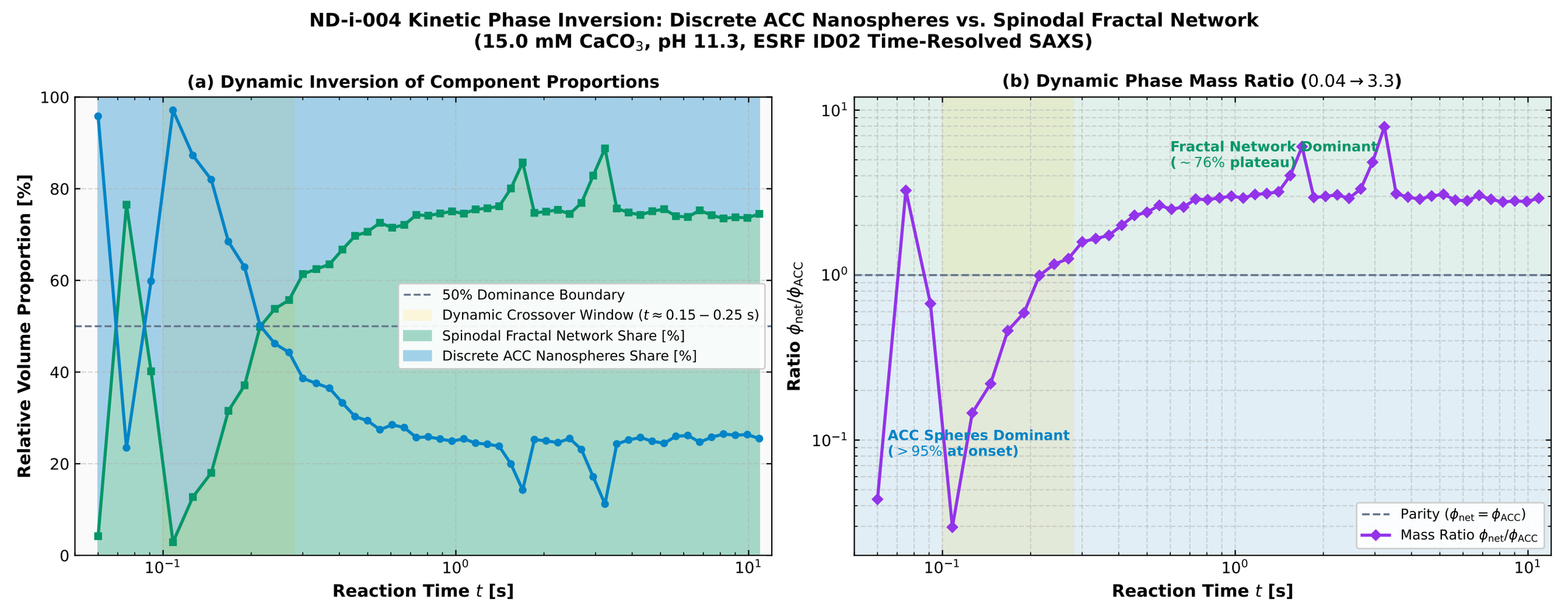
ND-i-004 Kinetic Phase Inversion: Discrete ACC Nanospheres vs. Spinodal Fractal Network
(15.0 mM CaCO3, pH 11.3, ESRF ID02 Time-Resolved SAXS)
(a) Dynamic Inversion of Component Proportions
Relative Volume Proportion [%]
Reaction Time t [s]
50% Dominance Boundary
Dynamic Crossover Window (t ≈ 0.15 − 0.25 s)
Spinodal Fractal Network Share [%]
Discrete ACC Nanospheres Share [%]
(b) Dynamic Phase Mass Ratio (0.04 → 3.3)
Ratio ϕnet/ϕACC
Reaction Time t [s]
Fractal Network Dominant
(~76% plateau)
ACC Spheres Dominant
(>95% at onset)
Parity (ϕnet = ϕACC)
Mass Ratio ϕnet/ϕACC